AI Reporting Agent — System Architecture

User (browser)

prompt (SSE)

Agent UI — ui.py (host :8501, FastAPI + SSE)

Schema tab
/discovery/* endpoints
browse · run · describe

ReAct agent (LangGraph)
LLM
one model: CLAUDE_MODEL (default Haiku)
/chat · /resume · /feedback

History summariser
LLM
compress old turns

Checkpointer
Postgres (CHECKPOINT_DB_URL)
or in-memory

traces

Langfuse (cloud)
traces, optional

tool call

Local tools
create_chart → Chart.js
edit_cube_config (form interrupt)

build_query
LLM
route to one view → structured query
validate vs schema · repair once

cube-mcp (:5001, SSE)
get_cube_metadata · reload_schema
query_cube + guard
(only fields from build_query)

library-mcp (:5002, SSE)
config CRUD · versions · rollback
save_graph · dashboards
commit → reload → auto-rollback

few-shot examples

/meta

Chart preview
/chart · /graph · /dashboard
replay.py (live re-query)

Feedback store
ratings + corrected queries
Postgres or SQLite

Cube.js (:4000)
semantic layer (views)
builds model from Library

reads

Library API (:3001)
versioned cube configs
graphs · dashboards

seed.py

Schema discovery
LLM
profile → grain → joins
+ AI descriptions

live re-query

SQL

SQL

Postgres data-db (:5432)
reporting data (per-dataset schemas)

Postgres library-db (:5433)
configs · graphs · dashboards

tables (publish.py)

Legend
- LLM call (red badge)
- entry / UI
- tool / code
- service
- datastore